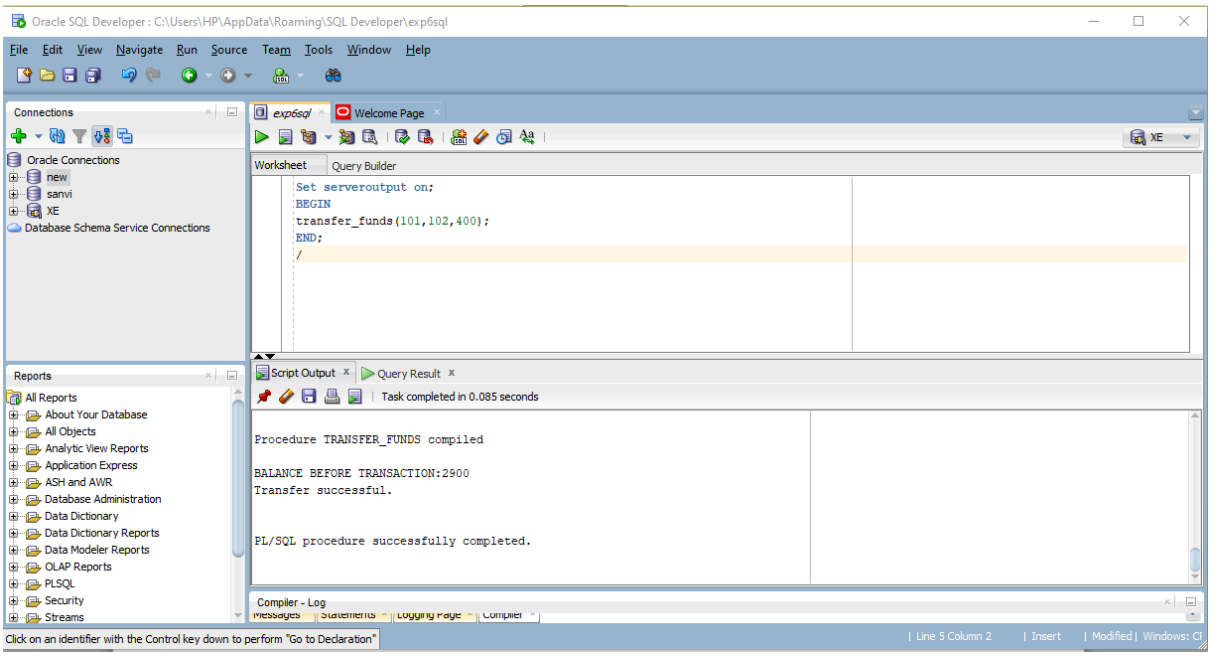
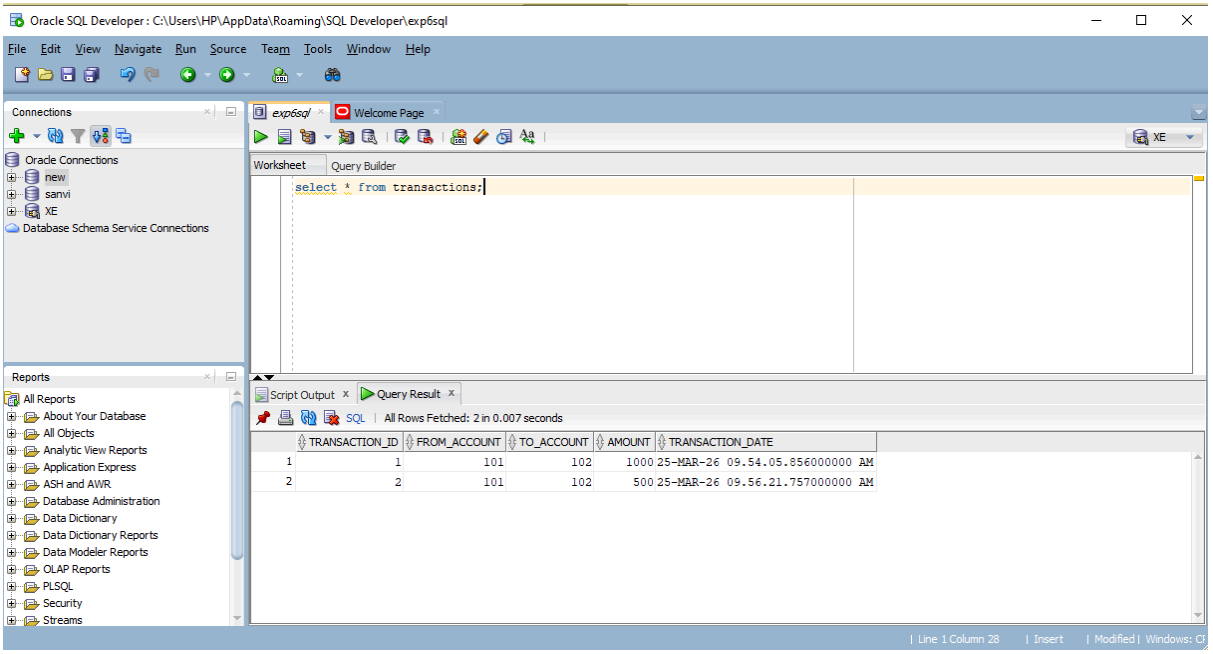
Practical 8 DBMS LAB:-

Name:- Sanvi Wadhankar
Sec:- C
Year :- 2nd
SEM:- 4TH

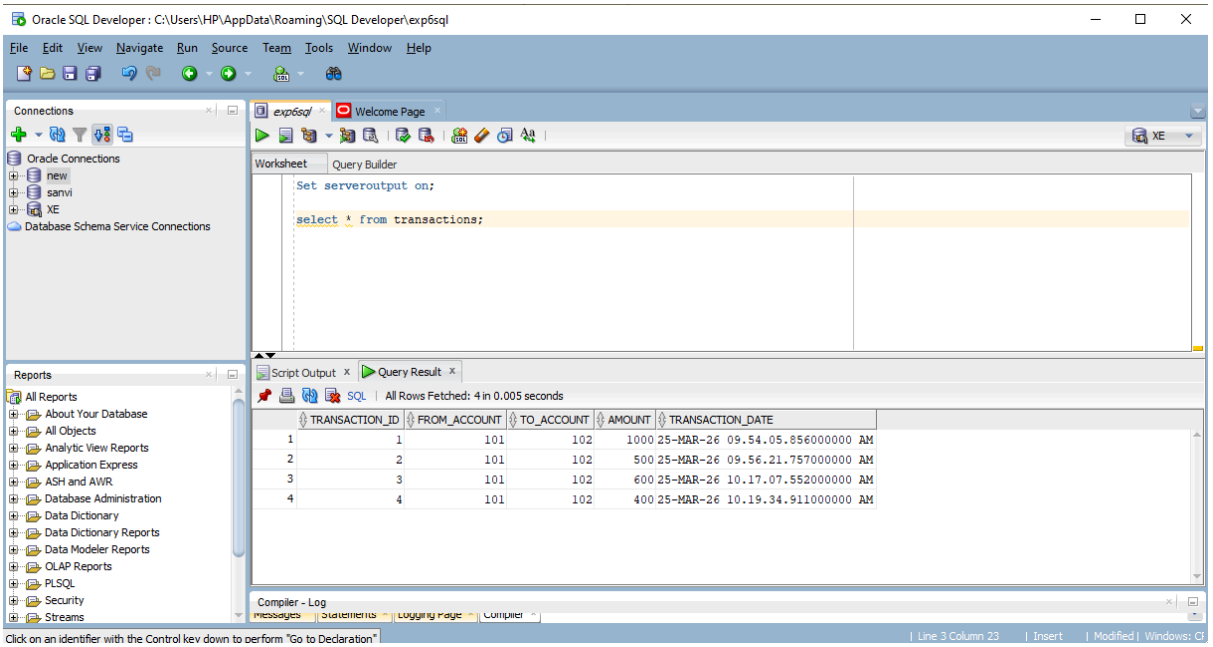
TASK 1:-



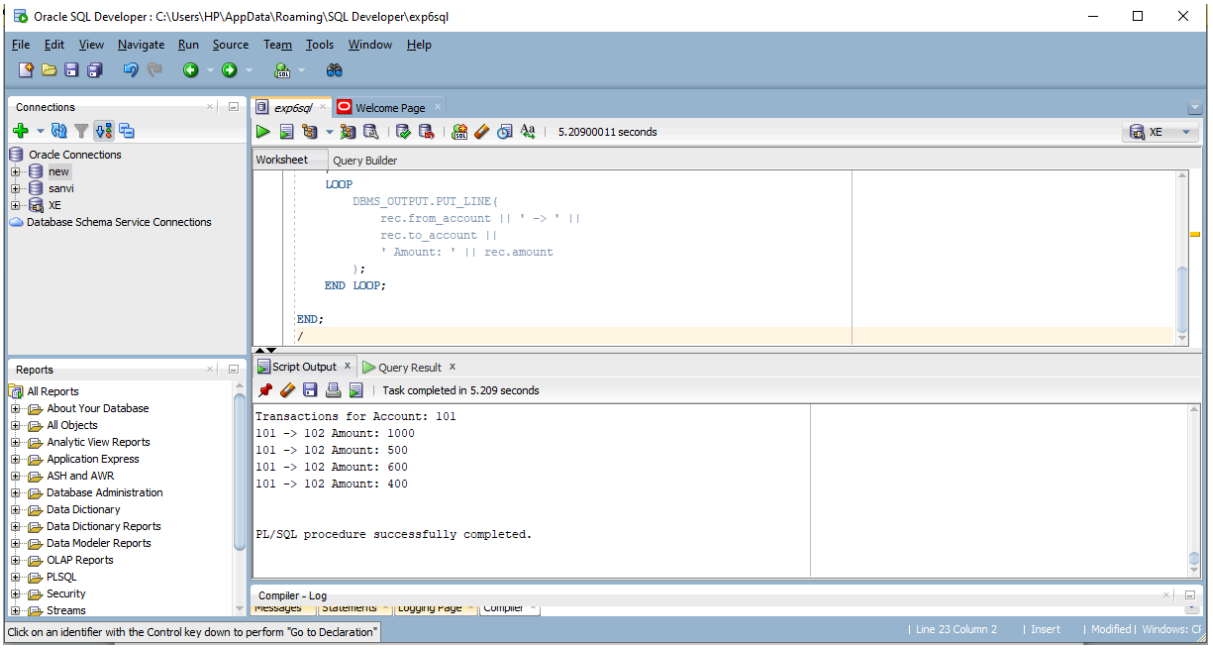
TASK 2:-



TASK 3:-



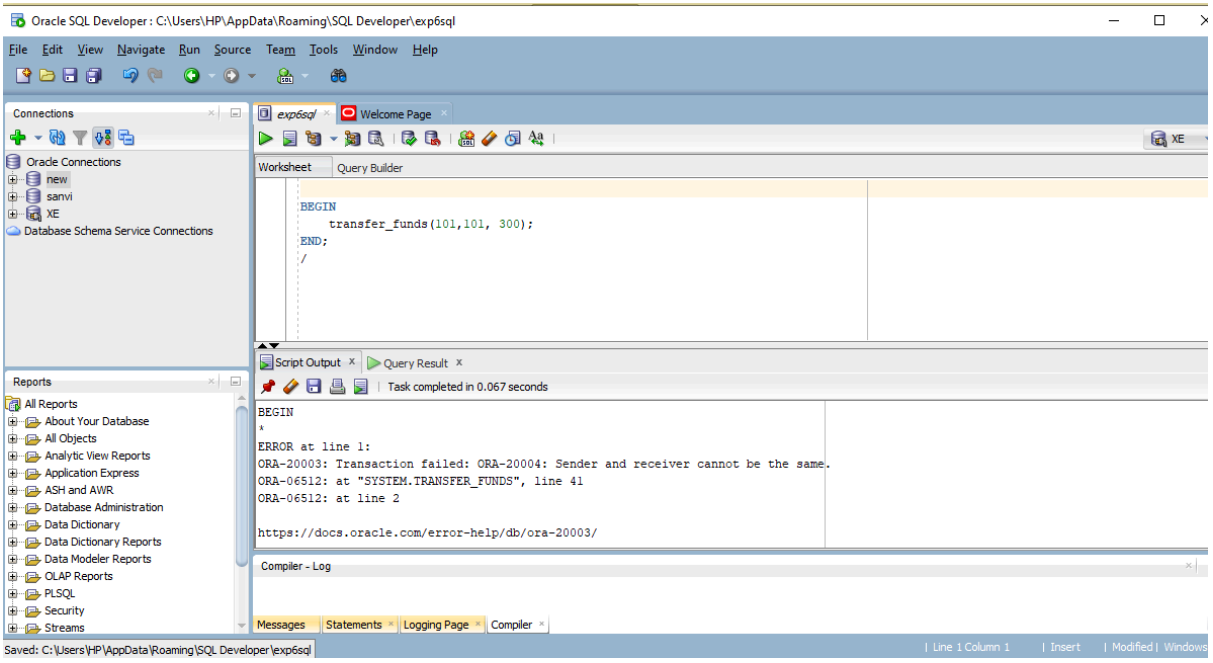
TASK 4:-



TASK 5:-

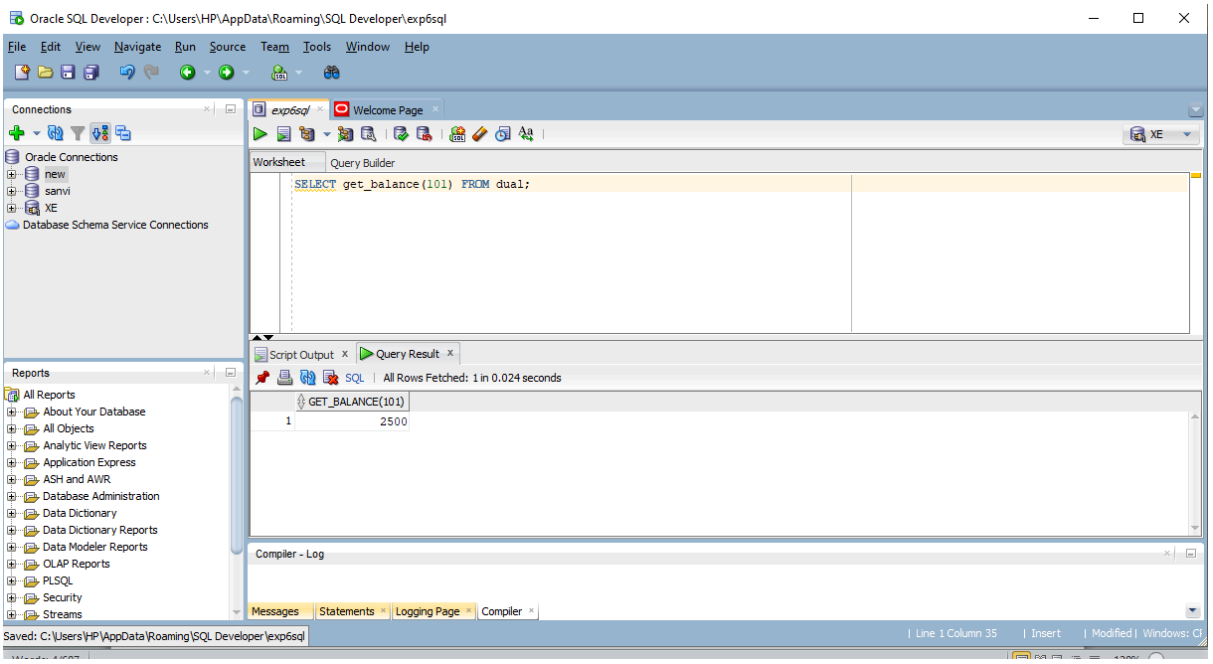
```
CREATE OR REPLACE PROCEDURE transfer_funds(
   p_from_acc NUMBER,
   p_to_acc NUMBER,
   p_amount NUMBER
```

```
) AS
   v_balance NUMBER;
BEGIN
   -- Check if sender has sufficient balance
   SELECT balance INTO v_balance FROM accounts WHERE account_no = p_from_acc;
   DBMS_OUTPUT.PUT_LINE('VALUE BEFORE TRANSACION' || v_balance);

   IF v_balance < p_amount THEN
      RAISE_APPLICATION_ERROR(-20001, 'Insufficient balance.');
   END IF;
   IF p_from_acc = p_to_acc THEN
   RAISE_APPLICATION_ERROR(-20004, 'Sender and receiver cannot be the same.');
   END IF;
   -- Deduct amount from sender
   UPDATE accounts SET balance = balance - p_amount WHERE account_no =
p_from_acc;

   -- Add amount to receiver
   UPDATE accounts SET balance = balance + p_amount WHERE account_no = p_to_acc;

   -- Log transaction
   INSERT INTO transactions (from_account, to_account, amount)
   VALUES (p_from_acc, p_to_acc, p_amount);

   -- Commit transaction
   COMMIT;

   DBMS_OUTPUT.PUT_LINE('Transfer successful.');



EXCEPTION
   WHEN NO_DATA_FOUND THEN
      RAISE_APPLICATION_ERROR(-20002, 'Invalid account number.');
   WHEN OTHERS THEN
      ROLLBACK;
      RAISE_APPLICATION_ERROR(-20003, 'Transaction failed: ' || SQLERRM);
END;
/
```



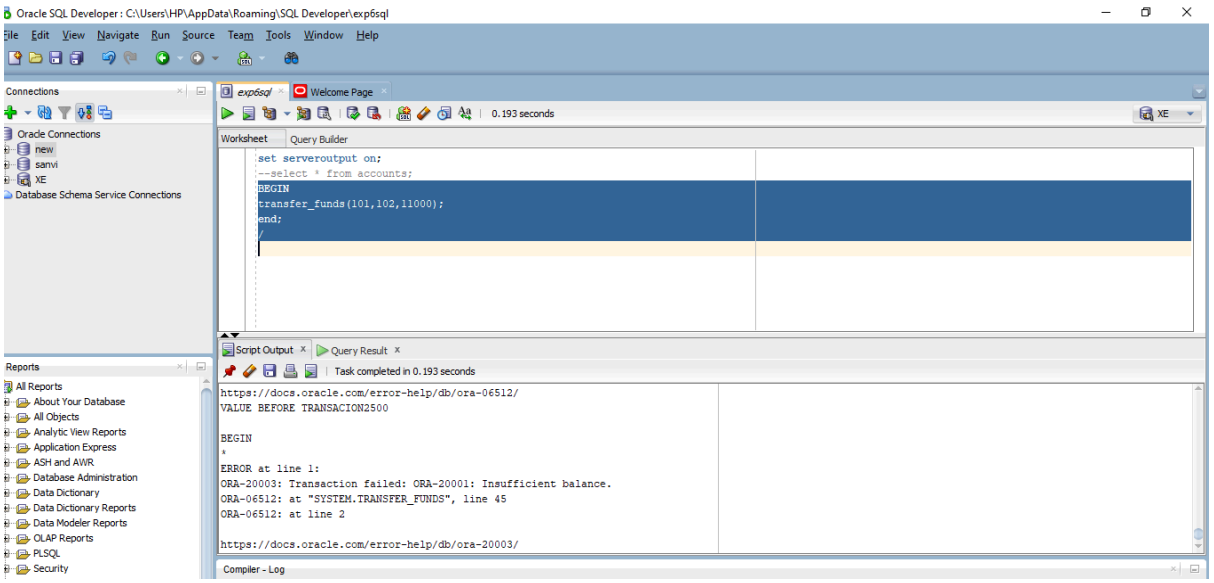
TASK 6:-

```
CREATE OR REPLACE FUNCTION get_balance(p_acc_no NUMBER) RETURN
NUMBER AS

      v_balance NUMBER;

BEGIN

      SELECT balance INTO v_balance FROM accounts WHERE account_no =
p_acc_no;

      RETURN v_balance;

END;

/
```



TASK 7:-

```
SET SERVEROUTPUT ON;
CREATE OR REPLACE PROCEDURE transfer_funds(
   p_from_acc NUMBER,
   p_to_acc NUMBER,
   p_amount NUMBER
) AS
   v_balance NUMBER;
BEGIN
   -- Check if sender has sufficient balance
   SELECT balance INTO v_balance FROM accounts WHERE account_no = p_from_acc;
   DBMS_OUTPUT.PUT_LINE('VALUE BEFORE TRANSACION' || v_balance);

   IF v_balance < p_amount THEN
      RAISE_APPLICATION_ERROR(-20001, 'Insufficient balance.');
   END IF;
   IF p_from_acc = p_to_acc THEN
   RAISE_APPLICATION_ERROR(-20004, 'Sender and receiver cannot be the same.');
   END IF;

   IF P_AMOUNT > 10000 THEN
   RAISE_APPLICATION_ERROR(-20005, 'Transfer amount exceeds the limit of ₹10,000.');
   END IF;
   -- Deduct amount from sender
   UPDATE accounts SET balance = balance - p_amount WHERE account_no =
p_from_acc;
```

```
    -- Add amount to receiver
    UPDATE accounts SET balance = balance + p_amount WHERE account_no = p_to_acc;

    -- Log transaction
    INSERT INTO transactions (from_account, to_account, amount)
    VALUES (p_from_acc, p_to_acc, p_amount);

    -- Commit transaction
    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Transfer successful.');



EXCEPTION
    WHEN NO_DATA_FOUND THEN
        RAISE_APPLICATION_ERROR(-20002, 'Invalid account number.');
    WHEN OTHERS THEN
        ROLLBACK;
        RAISE_APPLICATION_ERROR(-20003, 'Transaction failed: ' || SQLERRM);
END;
/
```

Task 8 ➖